
DENNIS GUNAWAN



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IF351 DATABASE SYSTEMS

01 INTRODUCTION TO DATABASE SYSTEM



COURSE SUB LEARNING OUTCOMES (SUB-CLO)

- Sub-CLO 1
 - Students are able to explain database, database users, and database system concept and architecture (C2)

OUTLINE

- Database
- Database Management System (DBMS)
- Characteristics of the Database Approach
- Actors on the Scene
- Advantages of Using the DBMS Approach
- When Not to Use a DBMS
- Data Model
- Three-Schema Architecture
- Data Independence
- DBMS Languages
- Database System Environment
- Architectures for DBMSs

INTRODUCTION

- Database and database systems are an essential component of everyday life in modern society
 - Deposit / withdraw funds in a bank
 - Make a hotel / airline reservation
 - Purchase something online
- Traditional database applications
 - The information that is stored and accessed is either textual or numeric

DATABASE

A collection of **related data**

- A database represents some aspect of the real world, sometimes called the **miniworld** or the **universe of discourse**
- Changes to the miniworld are reflected in the database
- A database is designed, built, and populated with data for a **specific purpose**

REAL WORLD & MINIWORLD

- Real world: real things in the world



- Miniworld: abstraction of the real world

Brand	Model	Color
Ford	Fiesta	Mars Red



Brand	Model	Color
Ford	Ecosport	Mars Red

- Changes must be reflected in the database as soon as possible

DATABASE MANAGEMENT SYSTEM (DBMS)

A collection of programs
that enable users
to create and maintain a database

A general-purpose software system
that facilitates the processes of
defining, constructing, manipulating, and sharing
databases among various users and applications

DATABASE MANAGEMENT SYSTEM (DBMS)

- Defining a database involves specifying the data types, structures, and constraints of the data to be stored in the database
- The database definition or descriptive information is also stored in the database in the form of a database catalog or dictionary
 - It is called **meta-data**
 - The information of data table is one kind of meta-data

DATABASE MANAGEMENT SYSTEM (DBMS)

- Constructing the database is the process of storing the data on some storage medium that is controlled by the DBMS
- Manipulating a database includes functions such as querying the database to retrieve specific data, updating the database to reflect changes in the miniworld
- Sharing a database allows multiple users and programs to access the database simultaneously

EXAMPLE

- A **university** database for maintaining information concerning students, courses, and grades in a university environment

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

CHARACTERISTICS OF THE DATABASE APPROACH

- A number of characteristics distinguish the database approach from the traditional approach of programming with files
 - Self-describing nature of a database system
 - Insulation between programs and data, and data abstraction
 - Support of multiple views of the data
 - Sharing of data and multiuser transaction processing

SELF-DESCRIBING NATURE OF A DATABASE SYSTEM

- Database system contains not only the database itself but also a complete definition or description of the database structure and constraints
 - This definition is stored in the DBMS catalog
 - The information stored in the catalog is called **meta-data**
- The catalog is used by the DBMS software and also by database users who need information about the database structure

RELATIONS

Relation_name	No_of_columns
STUDENT	4
COURSE	4
SECTION	5
GRADE_REPORT	3
PREREQUISITE	2

COLUMNS

Column_name	Data_type	Belongs_to_relation
Name	Character (30)	STUDENT
Student_number	Character (4)	STUDENT
Class	Integer (1)	STUDENT
Major	Major_type	STUDENT
Course_name	Character (10)	COURSE
Course_number	XXXXNNNN	COURSE
....		
....		
....		
Prerequisite_number	XXXXNNNN	PREREQUISITE

SELF-DESCRIBING NATURE OF A DATABASE SYSTEM

- In traditional file processing, data definition is typically part of the application programs themselves
- The data structure is declared in the application programs
- For example, an application program written in C++ may have struct or class declarations

INSULATION BETWEEN PROGRAMS AND DATA & DATA ABSTRACTION

- In traditional file processing, the structure of data files is embedded in the application programs
 - Any changes to the structure of a file may require changing all programs that access that file
- DBMS access programs do not require such changes in most cases
 - The structure of data files is stored in the DBMS catalog separately from the access programs
 - **Program-data independence**

Data Item Name	Starting Position in Record	Length in Characters (bytes)
Name	1	30
Student_number	31	4
Class	35	1
Major	36	4

- What if we add a new data item Birth_date?
 - In traditional file processing approach
 - In database approach

SUPPORT OF MULTIPLE VIEWS OF THE DATA

- A database typically has many users, each of whom may require a different perspective or view of the database
- A **view** may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored

SUPPORT OF MULTIPLE VIEWS OF THE DATA

- One user of the database may be interested only in accessing and printing the transcript of each student

TRANSCRIPT

Student_name	Student_transcript				
	Course_number	Grade	Semester	Year	Section_id
Smith	CS1310	C	Fall	08	119
	MATH2410	B	Fall	08	112
Brown	MATH2410	A	Fall	07	85
	CS1310	A	Fall	07	92
	CS3320	B	Spring	08	102
	CS3380	A	Fall	08	135

- A second user is interested only in checking that students have taken all the prerequisites of each course for which they register

COURSE_PREREQUISITES

Course_name	Course_number	Prerequisites
Database	CS3380	CS3320
		MATH2410
Data Structures	CS3320	CS1310

SHARING OF DATA & MULTIUSER TRANSACTION PROCESSING

- A multiuser DBMS must allow multiple users to access the database at the same time
- The DBMS must include **concurrency control** software to ensure that several users trying to update the same data do so in a controlled manner so that the result of the updates is correct
 - For example, when several reservation clerks try to assign a seat on an airline flight, the DBMS should ensure that each seat can be accessed by only one clerk at a time for assignment to a passenger
- A **transaction** is an executing program or process that includes one or more database accesses, such as reading or updating of database records
 - Either all the database operations in a transaction are executed or none are

ADVANTAGES OF USING THE DBMS APPROACH

- Controlling redundancy
- Restricting unauthorized access
- Providing storage structures for efficient query processing
- Providing backup and recovery
- Enforcing integrity constraints

CONTROLLING REDUNDANCY

- In traditional software development utilizing file processing, every user group maintains its own files for handling its data-processing applications
- Redundancy in storing the same data multiple times leads to several problems
 - **Duplication of effort:** the need to perform a single logical update multiple times (once for each file where student data is recorded)
 - **Storage space is wasted** when the same data is stored repeatedly
 - Files that represent the same data may become **inconsistent** because an update is applied to some of the files but not to others
 - Even if an update is applied to all the appropriate files, the data concerning the student may still be inconsistent because the updates are applied independently by each user group

RESTRICTING UNAUTHORIZED ACCESS

- When multiple users share a large database, it is likely that most users will not be authorized to access all information in the database
- Example
 - Who can access financial data?
 - Who is allowed to delete data?

PROVIDING STORAGE STRUCTURES FOR EFFICIENT QUERY PROCESSING

- **Index:** specialized data structures to speed up disk search for the desired records
 - Indexes are typically based on tree data structures or hash data structures, suitably modified for disk search
- **Buffering module:** maintains parts of the database in main memory buffers
 - In order to process the database records needed by a particular query, those records must be copied from disk to memory
- **Query processing and optimization** module of the DBMS is responsible for choosing an efficient query execution plan for each query based on the existing storage structures

PROVIDING BACKUP AND RECOVERY

- A DBMS must provide facilities for recovering from hardware or software failures
- The **backup and recovery subsystem** of the DBMS is responsible for recovery

What if the computer system fails
in the middle of a complex update transaction?

ENFORCING INTEGRITY CONSTRAINTS

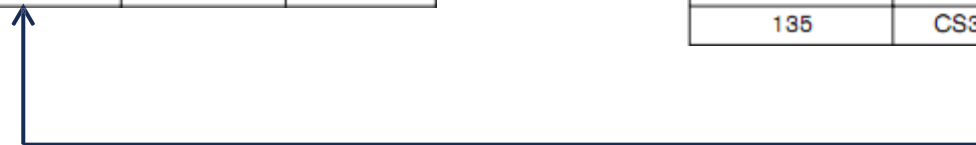
- **Data type constraint** specifies a data type for each data item
- **Referential integrity constraint** specifies that a record in one file must be related to records in other files
- **Key constraint** specifies uniqueness on data item values

COURSE

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WHEN NOT TO USE A DBMS

- Simple, well-defined database applications that are not expected to change
- Stringent, real-time requirements for some programs that may not be met because of DBMS overhead
- No multiple-user access to data

DATA MODEL

A collection of concepts that can be used to describe the structure of a database

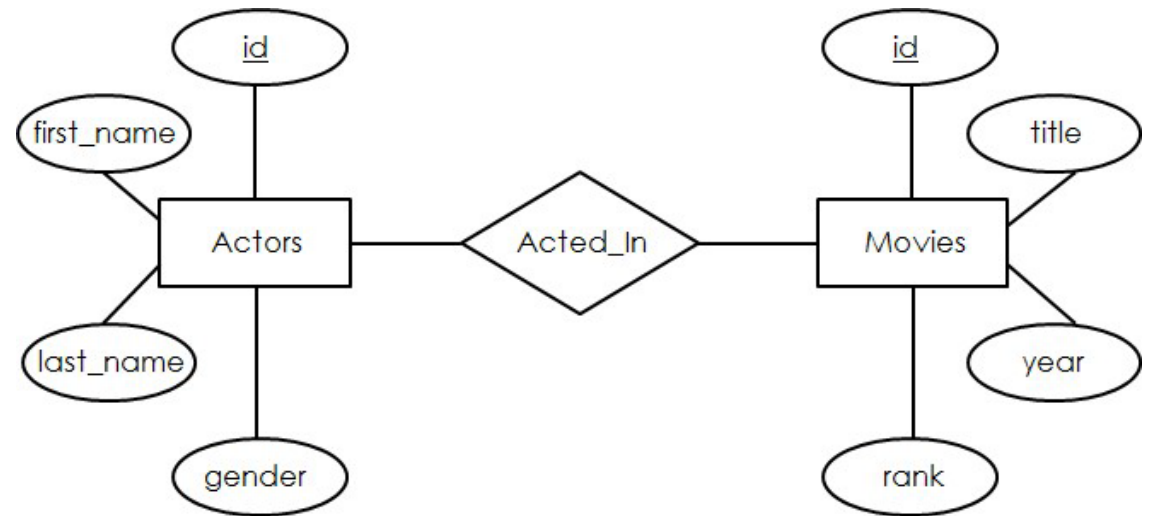
- Categories
 - High-level / conceptual data models
 - Representational / implementation data models
 - Low-level / physical data models

HIGH-LEVEL / CONCEPTUAL DATA MODELS

- Provide concepts that are close to the way many users perceive data
- Entity-Relationship model is a popular high-level conceptual data model

HIGH-LEVEL / CONCEPTUAL DATA MODELS

- **Entity** represents a real-world object or concept
- **Attribute** represents some property of interest that further describes an entity
- **Relationship** among two or more entities represents an association among two or more entities



REPRESENTATIONAL / IMPLEMENTATION DATA MODELS

- The models which are used most frequently in traditional commercial DBMSs
- Relational data model is widely used
- Represent data by using record structures and hence are sometimes called record-based data models

LOW-LEVEL / PHYSICAL DATA MODELS

- Describe how data is stored as files in the computer by representing information such as record formats, record orderings, and access paths
 - Access path: a structure that makes the search for particular database records efficient
 - Index is an example of an access path that allows direct access to data using an index term or a keyword

SCHEMAS

- It is important to distinguish between the description of the database and the database itself
- The description of a database is called the **database schema**
- Database schema is specified during database design and is not expected to change frequently

SCHEMAS

- A schema diagram displays only some aspects of a schema, such as the names of record types and data items, and some types of constraints

STUDENT

Name	Student_number	Class	Major
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schema construct

COURSE

Course_name	Course_number	Credit_hours	Department
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schema construct

PREREQUISITE

Course_number	Prerequisite_number
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schema construct

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
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schema construct

GRADE_REPORT

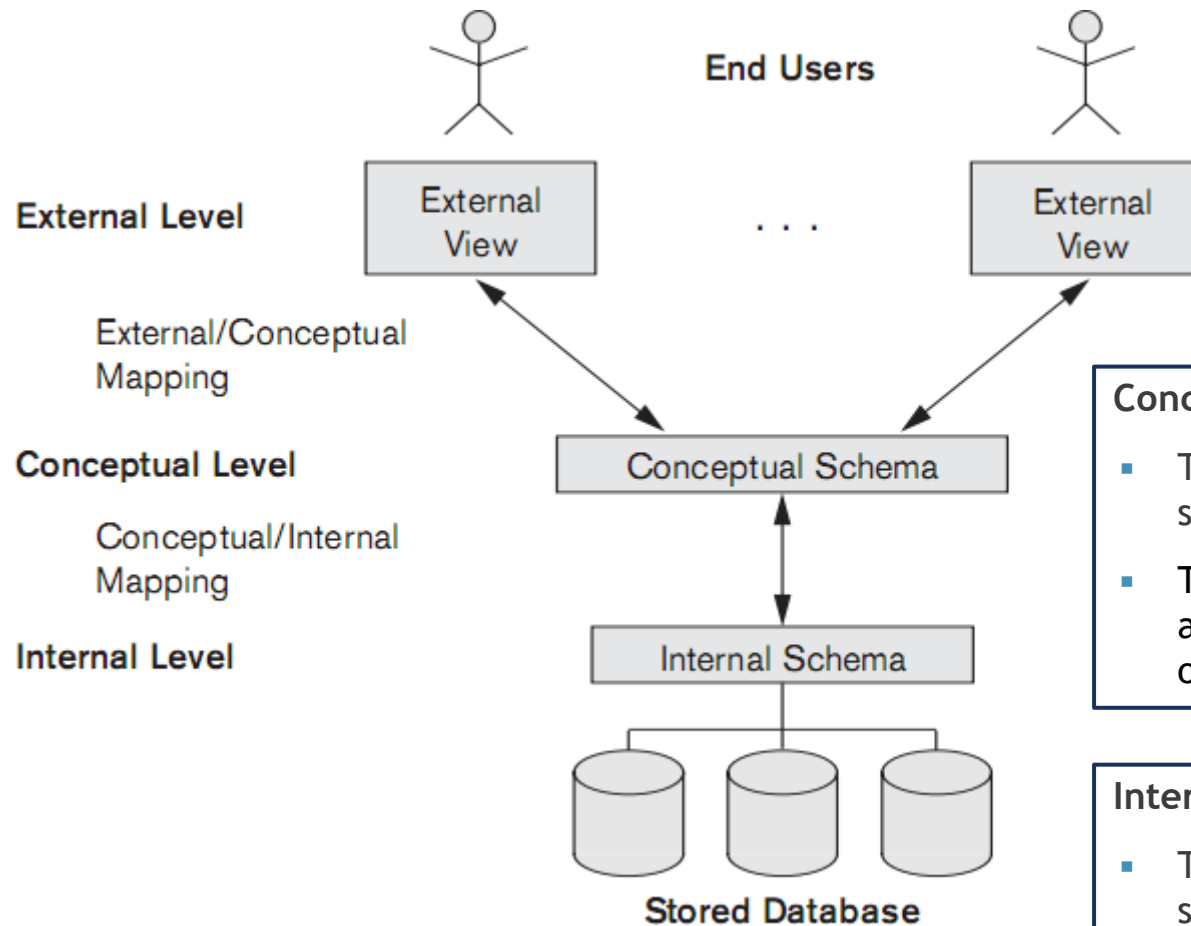
Student_number	Section_identifier	Grade
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schema construct

INSTANCES & DATABASE STATE

- The actual data in a database may change quite frequently
- The data in the database at a particular moment in time is called a **database state** or **snapshot**
- It is also called the **current set of occurrences** or **instances** in the database

THREE-SCHEMA ARCHITECTURE



External Level

- The external or view level includes a number of external schemas or user views
- Each external schema describes the part of the database that a particular user group is interested in and hides the rest of the database from that user group

Conceptual Level

- The conceptual level has a conceptual schema which describes the structure of the whole database for a community of users
- The conceptual schema hides the details of physical storage structures and concentrates on describing entities, data types, relationships, user operations, and constraints

Internal Level

- The internal level has an internal schema which describes the physical storage structure of the database

DATA INDEPENDENCE

The capacity to change the schema
at one level of a database system
without having to change the schema
at the next higher level

- Two types of data independence
 - Logical data independence
 - Physical data independence

LOGICAL DATA INDEPENDENCE

The capacity to change the conceptual schema
without having to change external schemas
or application programs

- Changing the conceptual schema to expand the database (by adding a record type or data item), to change constraints, or to reduce the database (by removing a record type or data item) without affecting the external schemas or application programs

PHYSICAL DATA INDEPENDENCE

The capacity to change the internal schema
without having to change the conceptual schema
&
the external schemas need not be changed as well

- Changes to the internal schema may be needed because some physical files were reorganized
- Creating additional access structures to improve the performance of retrieval or update

DATA INDEPENDENCE

- Physical data independence exists in most databases and file environments in which the physical details (exact location of data on disk, hardware details of storage encoding, etc.) are hidden from the user
- Logical data independence is very hard to come by because it allows structural and constraint changes without affecting application programs - a much stricter requirement

DBMS LANGUAGES

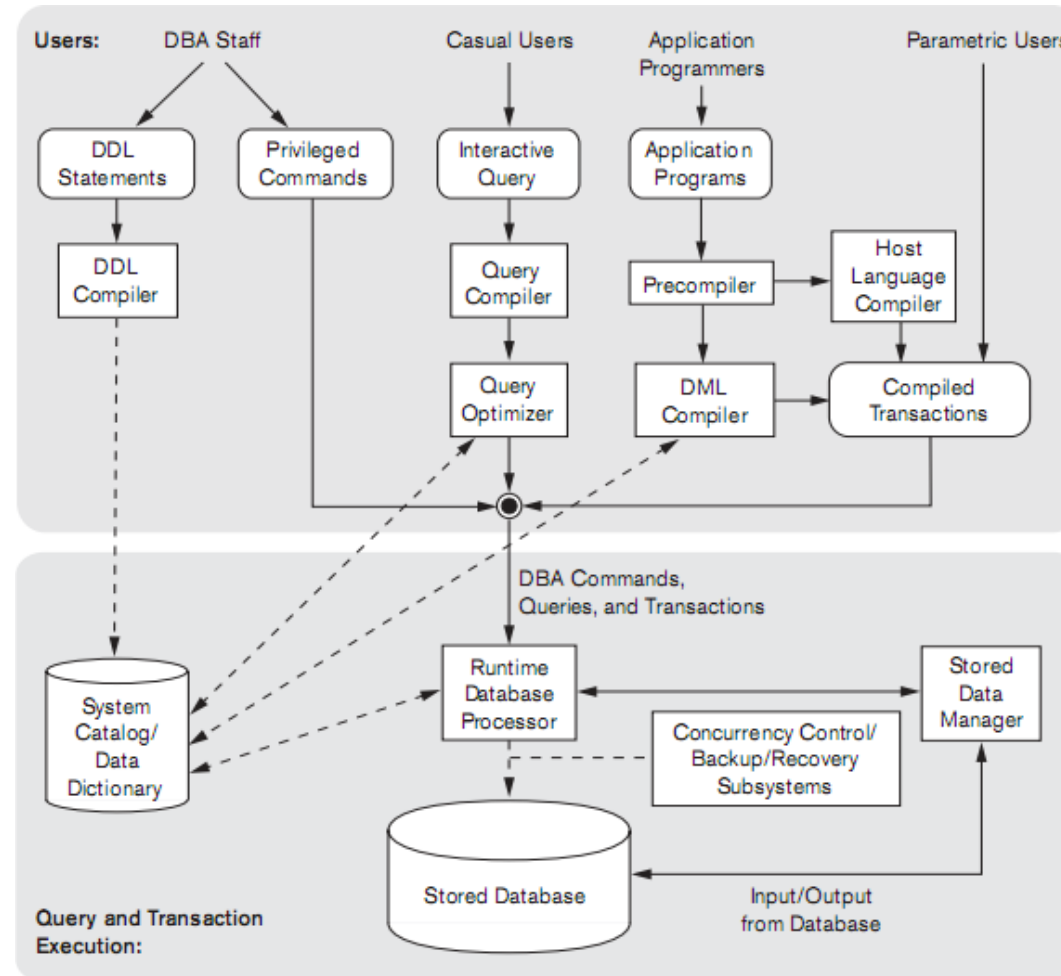
Data Definition Language (DDL)

- Is used by DBA and database designers to define conceptual and internal schemas
- The DBMS will have a DDL compiler whose function is to process DDL statements in order to identify descriptions of the schema constructs and to store the schema description in the DBMS catalog
- **CREATE, ALTER, TRUNCATE, DROP**

Data Manipulation Language (DML)

- Typical manipulations include retrieval, insertion, deletion, and modification of the data
- **SELECT, INSERT, UPDATE, DELETE**

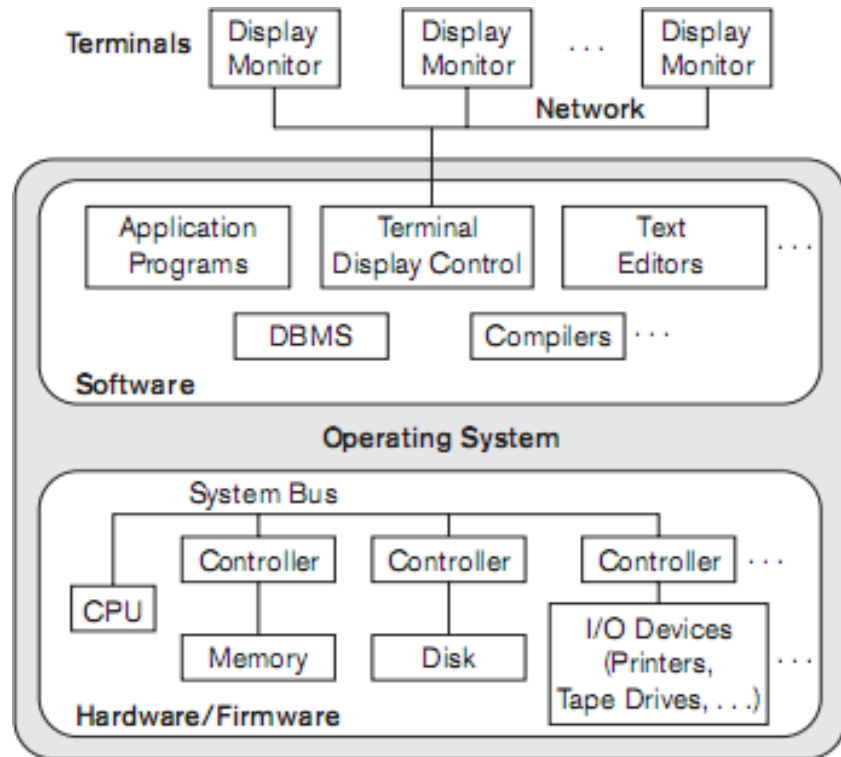
DBMS COMPONENT MODULES



DATABASE SYSTEM UTILITIES

- **Loading utility** is used to load existing data files into the database
- **Backup utility** creates a backup copy of the database, usually by dumping the entire database onto tape
- **Database storage reorganization utility** can be used to reorganize a set of database files into a different file organization to improve performance
- **Performance monitoring utility** monitors database usage and provides statistics to the DBA

CENTRALIZED DBMS ARCHITECTURE



- Most users accessed the system via computer terminals that did not have processing power and only provided display capabilities
- All processing was performed remotely on the computer system

TWO-TIER CLIENT / SERVER ARCHITECTURE

- In relational DBMS (RDBMS), many of which started as centralized systems, the system components that were first moved to the client side were the user interface and application programs
 - Client side: user interface programs and application programs (business logic and computing)
 - Server side: query and transaction functionality related to SQL processing (data access)

TWO-TIER CLIENT / SERVER ARCHITECTURE

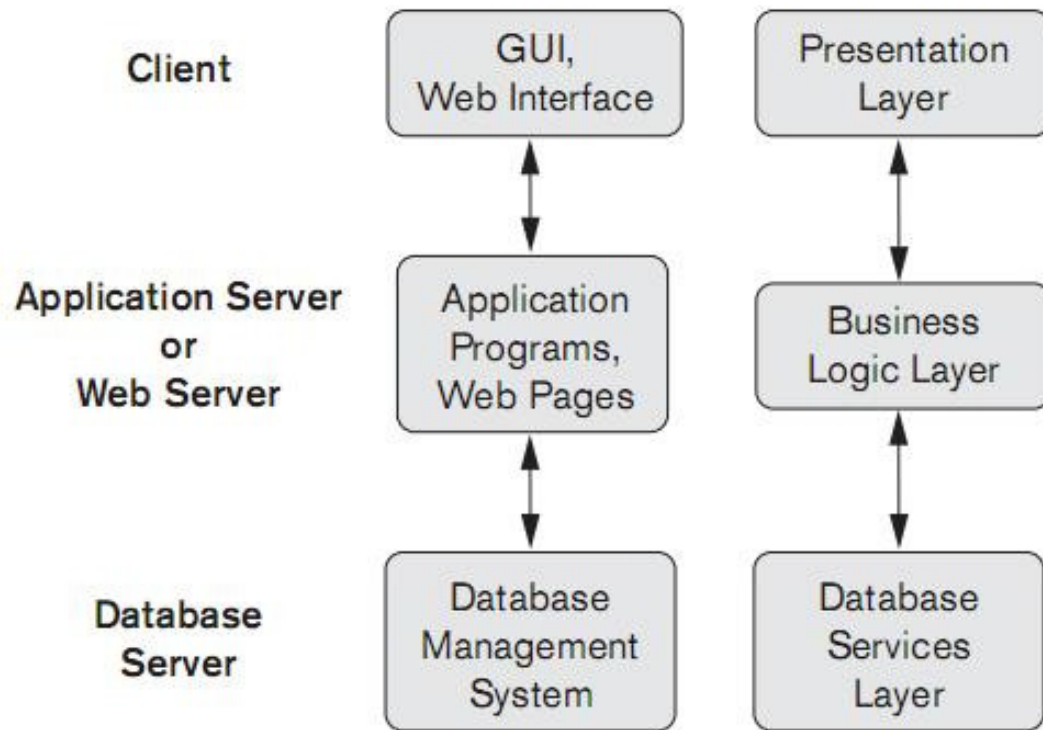
- The program is executed on client computer
- Client side and server side must install database middleware, such as ODBC (Open Database Connectivity), JDBC, or Oracle API
- Database middleware provides Application Programming Interface (API), which allows client-side programs to call the DBMS

Disadvantages

- Software deployment and update
- Middleware deployment and update



THREE-TIER ARCHITECTURE



- Intermediate layer or middle tier is sometimes called the application server and sometimes the Web server, depending on the application
- This server plays an intermediary role by running application programs and storing business rules (procedures or constraints) that are used to access data from the database server
- Middle tier can also improve database security by checking a client's credentials before forwarding a request to the database server

REFERENCES

- Elmasri, Ramez and Shamkant B. Navathe (2016), Fundamentals of Database Systems, 7th edition, Addison Wesley.
- Rick F.van der Lans (2007), SQL for MySQL Developers, Addison Wesley.
- Connolly, Thomas and Carolyn Begg (2015), Database Systems A Practical Approach to Design, Implementation, and Management, 6th edition, Pearson Education, ISBN 13: 978-1-292-06118-4, Essex, England.

NEXT WEEK: DATA MODELING USING THE ENTITY-RELATIONSHIP MODEL

- Database Design
- Entities
- Attributes
- Relationships
- ER Diagrams Notation

```
SELECT *  
FROM WORLD  
WHERE SOMEONE  
LIKE '%YOU%';
```

```
EMPTY SET (0.00 SEC)
```

Vision

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.



Mission

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology